

Exploratory Data Analysis

Naukri Data Analyst & Data Scientist Job Posting in India

Executive Summary:

62,468 cleaned job postings analyzed across 3 different independently scraped Naukri.com searches. The strongest finding in the dataset is **compensation, not skill demand**. Postings classified as **Scientist** or **Hybrid** (both “Analyst” and “Scientist”) carry a disclosed salary of roughly **double** that of **Analyst** and **Other Data Role** postings, i.e., **19/20 LPA** vs **10 LPA**. This holds even though **SQL** and general data-analysis skills appear across all 4 groups almost equally. The pay gaps track the title, not the tool stack. Only **10%** of postings disclosed salary at all, so this finding rests on a real but limited sample of 4,822 postings and should be read as directional, not exact.

1. Data Overview and Sourcing:

Source: [Job Postings Dataset from Naukri.com](#) (Kaggle). Downloaded as 3 separate files, representing 3 independent scrapes rather than 1 dataset that spills into parts.

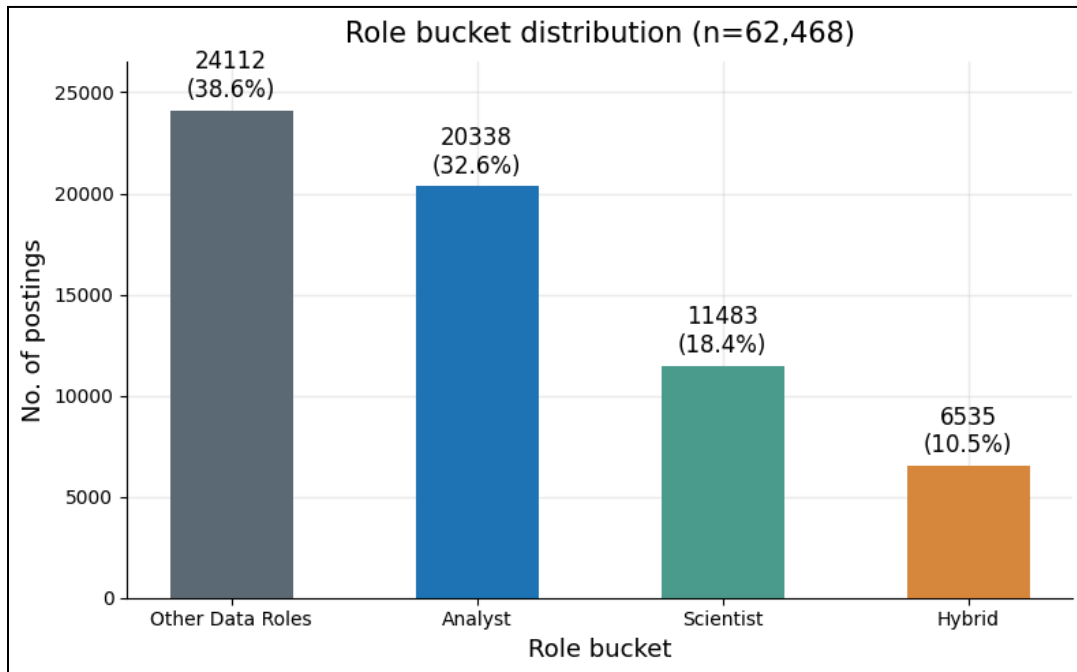
	Data Science.csv	Data Analytics.csv	combo.csv
Rows	47,191	20,064	19,649
Columns	8 (Has Post_Url & Post_Time)	8 (Has Post_Url & Post_Time)	6 (No Post_Url, No Post_Time)
Exact duplicate rows	24,399 (51.7%)	34 (0.17%)	3 (0.02%)
“Not disclosed” salary records	34,466 (73.04%)	18,145 (90.44%)	17,775 (90.46%)
Record after removing duplicates	22,792	20,030	19,646

2. Cleaning Steps Applied:

- Schema aligned across all 3 files (combo.csv column renamed to match). Post_Url and Post_Time left as null for combo.csv's **19,646 rows** rather than dropping them.
- Exact duplicates removed within each file (**24,436 rows**), then again after union/append across all 3.
- Text columns standardized by **lowercasing, trimming, and stripping** whitespace for consistent matching.
- Job titles classified into **4 buckets: Analyst / Scientist / Hybrid / Other Data Roles** using a title keyword calculated field.
- **Locations normalized** by standardizing city aliases and formatting variations for geographic analysis.
- Skills extracted via a **73-term** external vocabulary matched against the (undelimited, concatenated) skills text, since the raw field has no separate character between skills to delimit them.
- **Duplicate skills** within each posting **have been removed** so that a skill is counted at most once per job posting, preventing an inflation of skill-frequency statistics.
- Salary field evaluated from free text ("6-10 lacs pa") into a numeric LPA midpoint. **"Unpaid" posting flagged** and excluded from salary aggregates rather than silently averaged in.

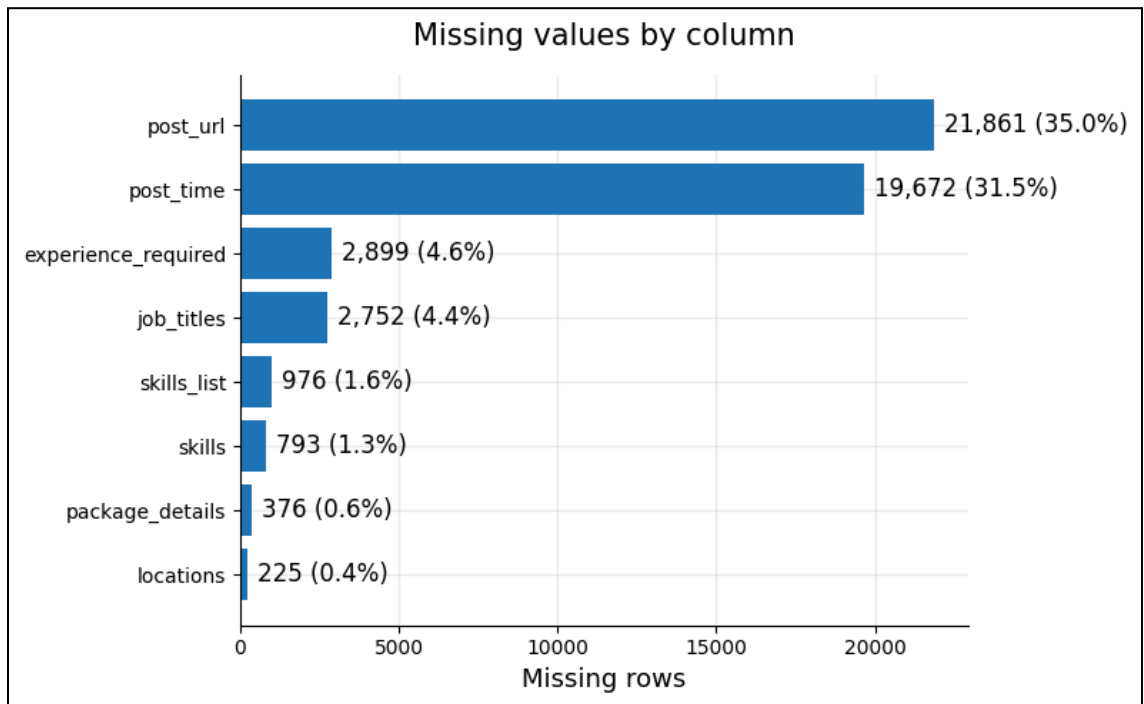
3. Univariate Analysis: (Single variable analysis)

- **Role Composition:**



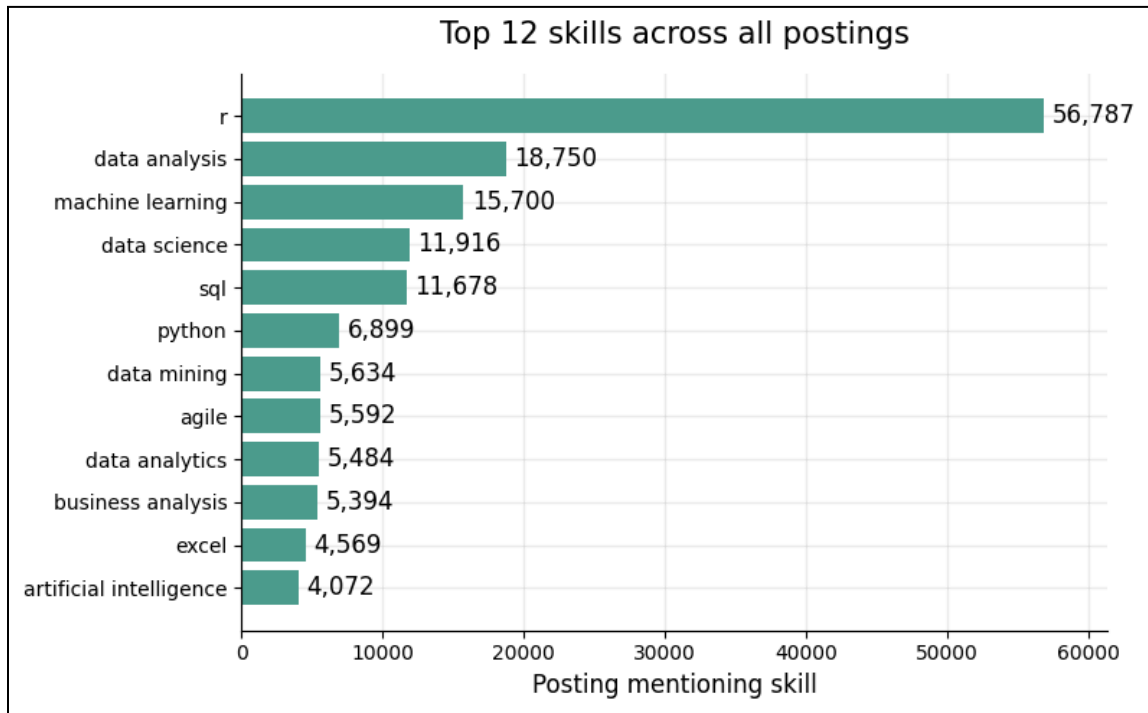
- The **“Other Data Roles”** category contains neither **“Analyst”** nor **“Scientist”**. It contains roles like *Data Governance Manager, Data Modeler, and similar*. It is the largest group at **38.6%**. This is because search terms are **not pre-filtered for Analyst & Scientist** roles.
-

- **Missing Values:**



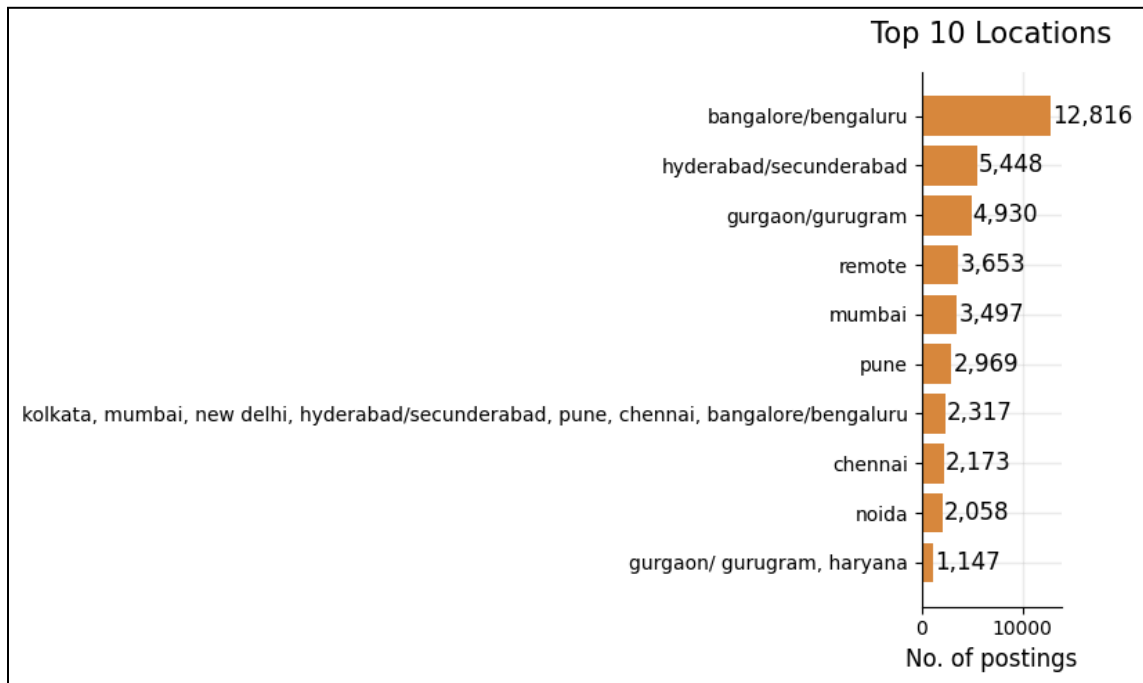
*Post_url and post_time missingness (35.0% / 31.5%) is structural and genuine. Because these columns simply don't exist in the combo.csv sheet with **19,646 rows**. It's not a data quality defect.*

- **Skill Demand:**



*An analysis of the top 12 skills across **62,468** job postings highlights 3 primary programming languages: **R, SQL, and Python**. Notably, the R language is specified in **56,787** job postings.*

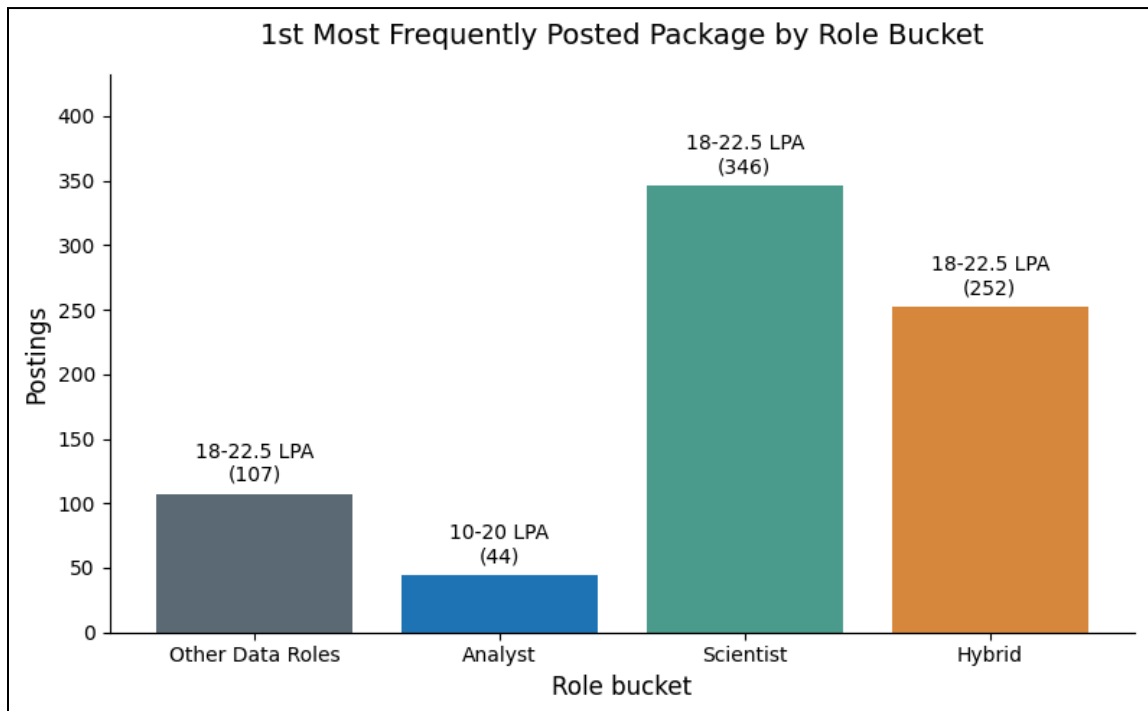
- **Location Concentration:**

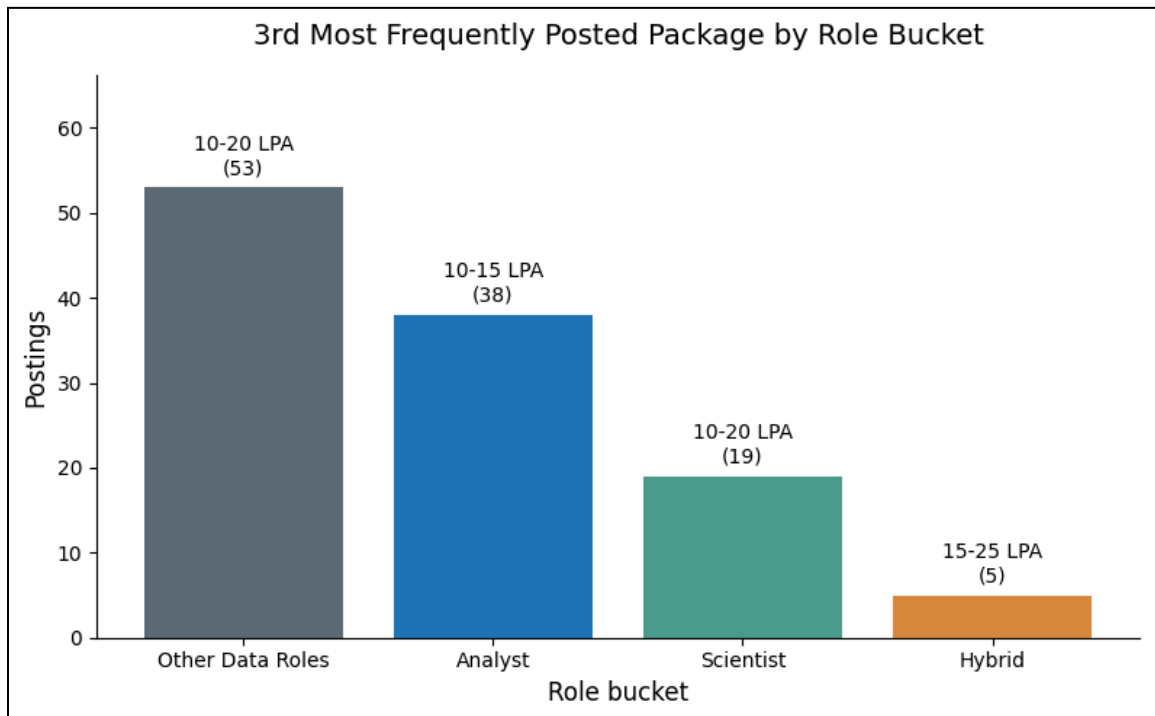
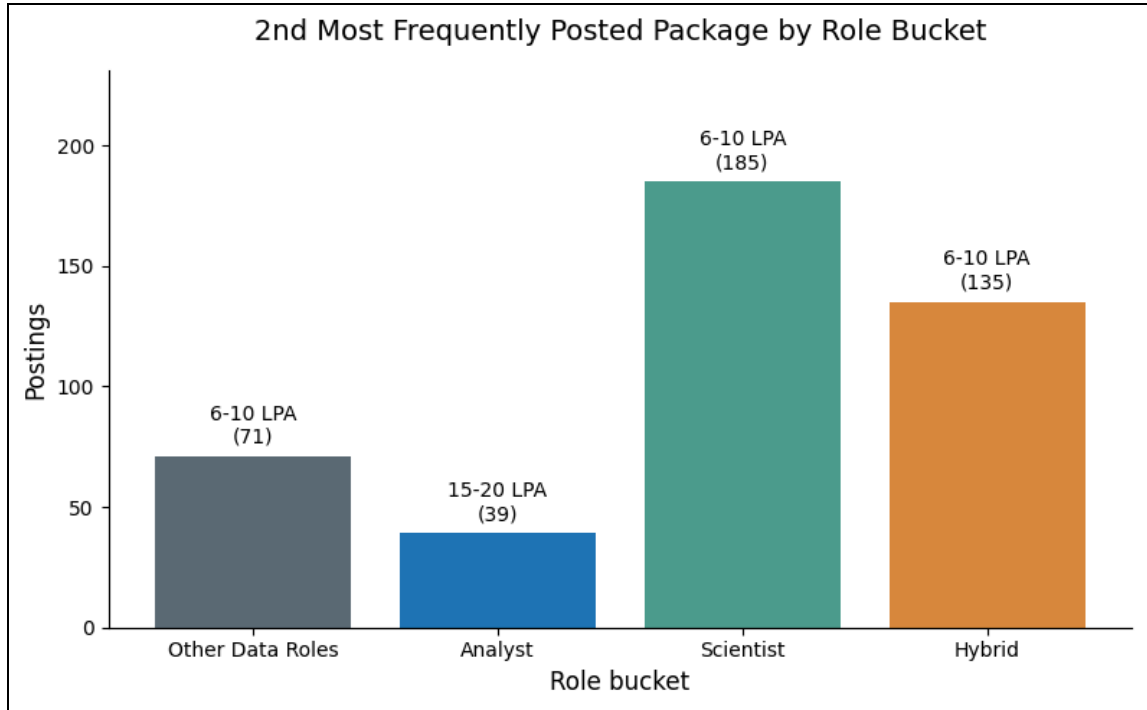


Bangalore/Bengaluru alone has more postings than the next 2 combined.
2,317 posting lists have multiple cities in a single cell. This may indicate flexibility in work location or that candidates across ***multiple cities are being targeted.***

4. Bivariate Analysis: (Two-variable analysis for relationship, correlation)

- **Salary by role bucket:**





*These package-frequency charts tell us which posted salary bands recur most often. **Scientist & Hybrid** postings have the **highest packages & high number of openings** compared to Analyst & Other Data Roles.*

- Summary of salary disclosed & not disclosed:

Role bucket	Total Postings	Not disclosed	Not disclosed %	Disclosed%
Other Data Roles	24,112	21,470	89.04%	10.95%
Analyst	20,338	18,573	91.32%	8.67%
Scientist	11,483	10,182	88.67%	11.32%
Hybrid	6,535	5,539	85.58%	15.24%

5. Top 5 Skills by Each Role Bucket with the count:

Analyst	Scientist	Hybrid	Other Data Roles
R (18,787)	R (10,215)	R (5,754)	R (22,031)
Data Analysis (4,671)	Machine Learning (5,830)	Machine Learning (4,003)	Data Analysis (8,279)
SQL (3,091)	Data Science (4,765)	Data Science (3,389)	Machine Learning (3,524)
Business Analysis (2,696)	Data Analysis (3,629)	Data Analysis (2,171)	SQL (3,283)
Machine Learning (2,343)	SQL (3,174)	SQL (2,130)	Data Science (2,658)

R, Data Analysis, and SQL are common across every role bucket, not the differences.
Business Analysis appears only in the Analyst top 5 skills but not in other role buckets.
Machine Learning is highly demanded for all roles except the Analyst role.

6. Key Findings:

- 1) **Data Scientist and Hybrid** are 2 roles for which we get a **high number of postings and packages** also, rather than Analyst and Other Data Roles.
 - Data Scientist: postings - **346 with 18-22.5 LPA**
 - Hybrid: postings - **252 with 18-22.5 LPA**
- 2) **R, Data Analysis, and SQL** appear in the top 5 skills for each role.
- 3) **Python** is not in the top 5 skills in these **62,468 postings**, which have a **count of 6,899**. Which is most important & widely used for all these 4 roles? Maybe this point requires consideration and research.
- 4) **On average, 88.65%** of all postings **do not disclose** the salary for all 4 roles.
- 5) **On average, only 11.54%** of postings **disclosed** the salary for all 4 roles.
- 6) **Bangalore/Bengaluru** exhibits the **highest geographic concentration**, outstripping the next 2 top locations combined in overall posting volume.
- 7) The “**Other Data Role**” category does not represent exceptional or outlier roles; out of a total of 62,468 job postings, **38.6% (24,112)** fall under this category.